

Unknown Title



 Description

Description

 Note

Note

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Submissions

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Code

Code



Testcase

Testcase

>_

Test Result

Test Result



Leet

Leet

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[20. Valid Parentheses](#)

Easy



Topics



Companies



Hint

Given a string `s` containing just the characters `'('`, `)'`, `'{'`, `'}'`, `'['` and `']'`, determine if the input string is valid.

An input string is valid if:

1. Open brackets must be closed by the same type of brackets.
2. Open brackets must be closed in the correct order.
3. Every close bracket has a corresponding open bracket of the same type.

Example 1:

Input: s = "()"

Output: true

Example 2:

Input: s = "()[]{}"

Output: true

Example 3:

Input: s = "["

Output: false

Example 4:

Input: s = "([])"

Output: true

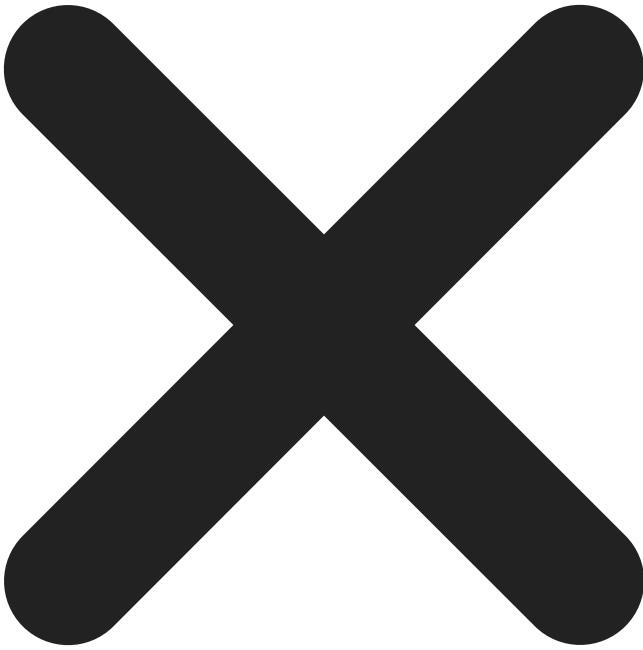
Example 5:

Input: s = "([)]"

Output: false

Constraints:

- $1 \leq s.length \leq 10^4$
- s consists of parentheses only '()[]{}'.



Seen this question in a real interview before?

1/6

Yes

No

Accepted

7,651,416/17.3M

Acceptance Rate

44.2%



Topics



[String](#) [Stack](#)



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Hint 1



Use a stack of characters.



Hint 2



When you encounter an opening bracket, push it to the top of the stack.



Hint 3



When you encounter a closing bracket, check if the top of the stack was the opening for it. If yes, pop it from the stack. Otherwise, return false.



Discussion (680)



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Discussion Rules



1. Please don't post **any solutions** in this discussion.
2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.
3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.



[sunwangshu](#)

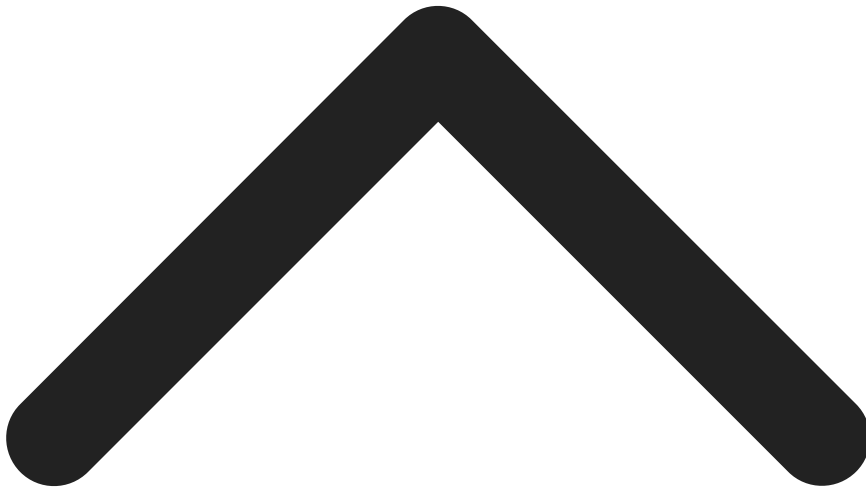
Aug 01, 2017

The brackets must close in the correct order, "()" and "()[]{}" are all valid but "]" and "[()]" are not.

From this problem description I cannot guess if `[]{}()` is valid or not.

But clearly from the answer it is valid.

Then I think the problem description should be updated.



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2.3K



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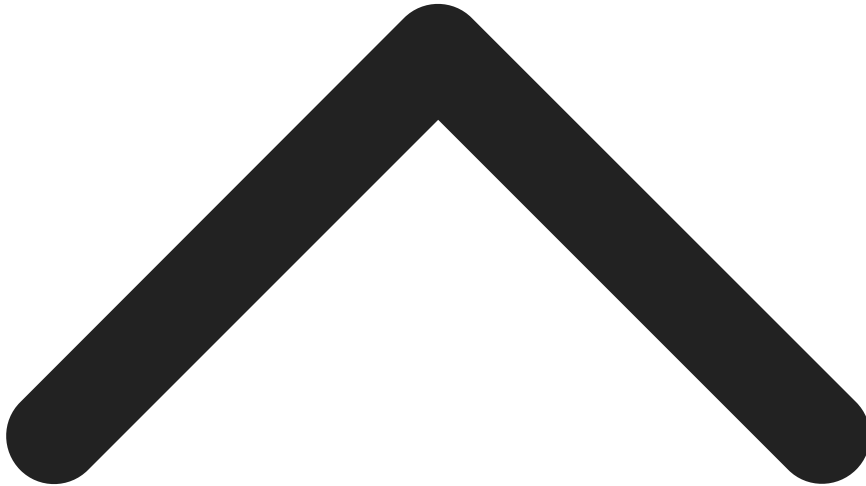


[bry.](#)



May 27, 2019

It's a really **good** problem. But I don't think it's appropriate to call it **Easy**. If you've never had a question like this before, and if you're new to python and programming - which is my guess the target audience for easy questions - it's a little difficult.



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739



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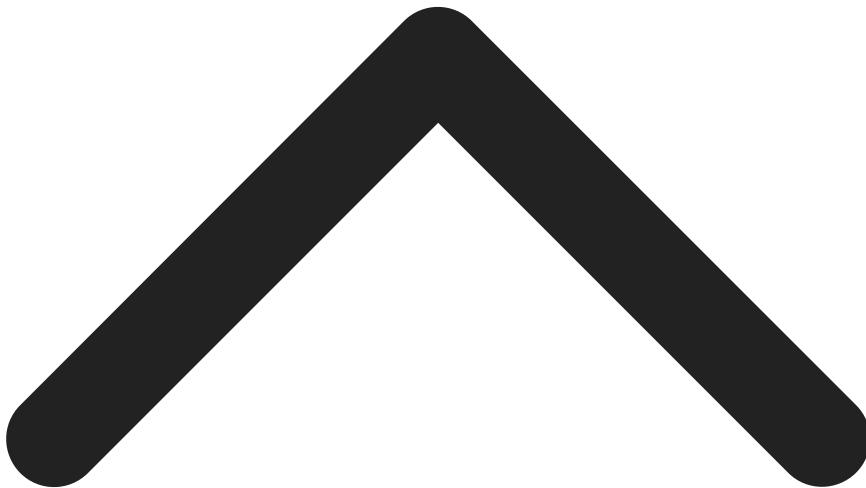
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[tonybrasunas](#)

Jun 05, 2017

The description for this one should say `([]{}))` is valid too. Otherwise it's not clear that validating nested parentheses is part of the requirements.



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793



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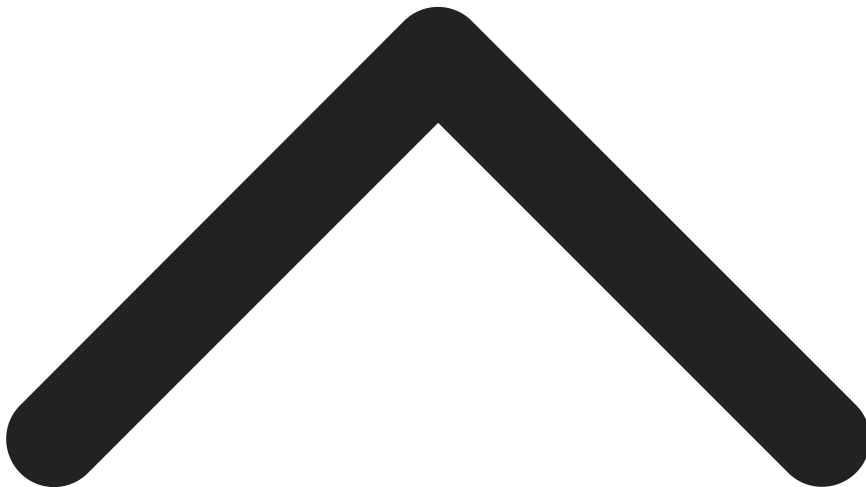


[Saksham Maitri](#)



Dec 03, 2023

either i am dumb af or this is not easy



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313



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[Alex](#)

Apr 27, 2023

Here is some tip:

The first line of the function uses a guard statement to check if the length of the input string is even. If the length is odd, the function returns false because it means that the brackets are not properly balanced.

The next line creates an empty stack of characters to store the opening brackets.

The function then loops through each character item in the input string s.

For each character item, the code checks whether it is an opening bracket: (, [, or {. If it is, the corresponding closing bracket is pushed onto the stack. For example, if the character is (, the character) is pushed onto the stack.

If the character is a closing bracket:),], or }, the code checks whether the stack is empty.

If the stack is empty, it means that there is no opening bracket to match the closing bracket, so the function returns false. If the stack is not empty, the last opening bracket is removed from the stack and checked against the current closing bracket. If the brackets do not match, the function returns false.

After looping through all the characters in the input string, the function checks whether the stack is empty. If it is, it means that all the opening brackets have been matched with their corresponding closing brackets, so the function returns true. If the stack is not empty, it means that there are unmatched opening brackets, so the function returns false.



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Tip



254



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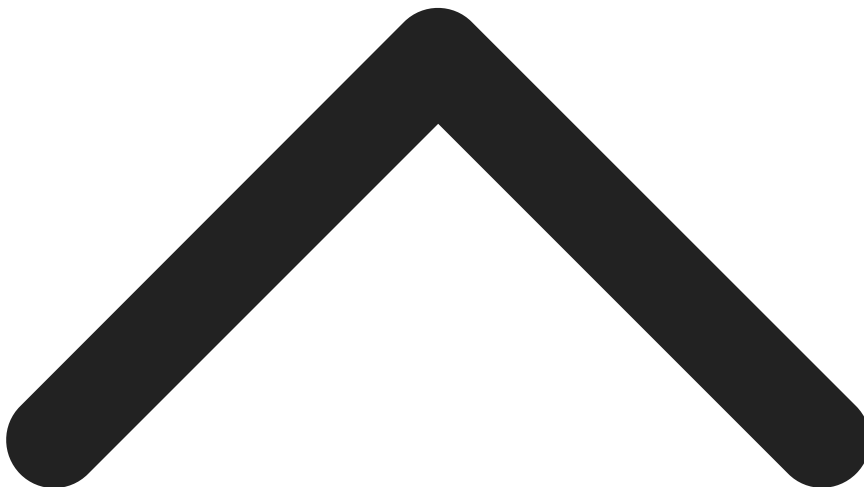
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[tryhard00](#)

Oct 13, 2022

This problem could be considered easy but only if you realized the crux of it. If you tried to do it using pure conditionals, which I did for the first 3 hours, you're in for a very bad time. I finally gave up on conditionals when it had a nested within nested case. I was about to go 3d array to store each set but it was so logically complex I almost quit. Then I saw it. So this problem could be considered medium if you didn't see the easy answer. Overall, a very clever question. It had so many caveats that I kept getting rejected and it was giving me pstd by the end. lol. For those who are stuck, the hint is what goes in last must come out first.



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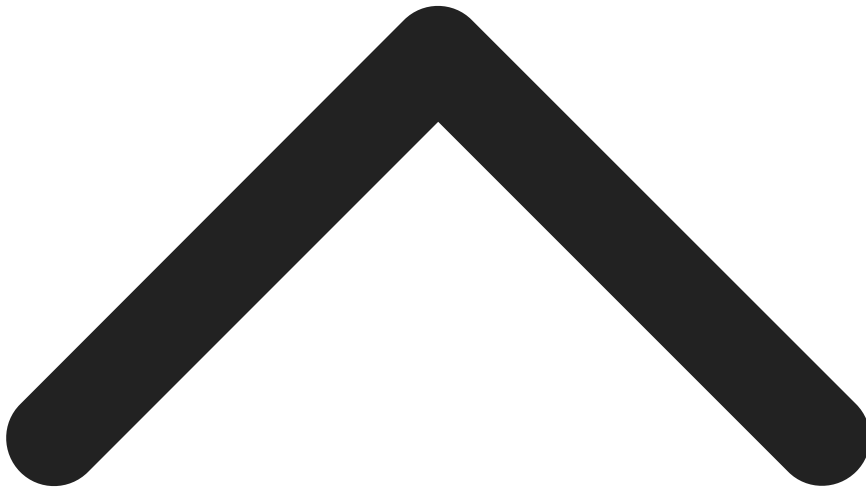
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[Alexander Akhilchenok](#)

Jun 05, 2023

Description is not good enough. Should be shown some more test cases as "([])", because "Open brackets must be closed in the correct order" could be misunderstood.



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[Ram](#)



Jan 01, 2021

whats wrong with this TC "{}" in custom test cases run it succeed but it fails on submission.

"



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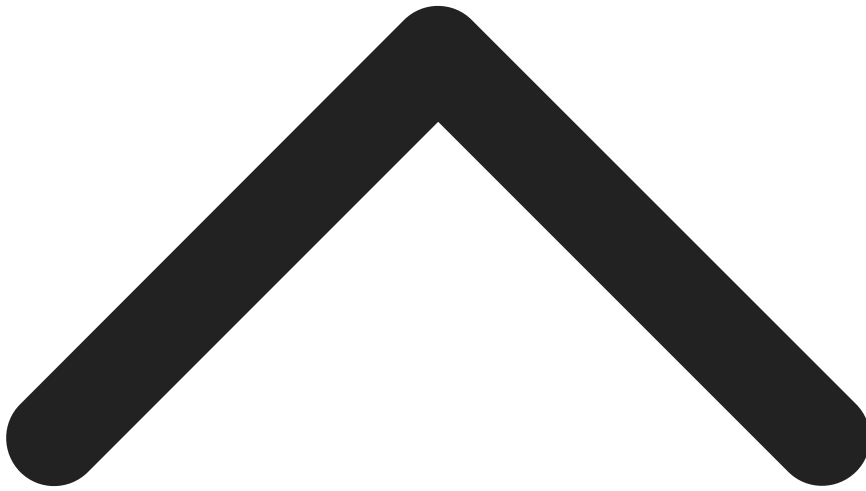
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[raja shekar davalgari](#)

Feb 09, 2020

for the string '()' my code is working fine with python 3.7, but here its showing wrong output



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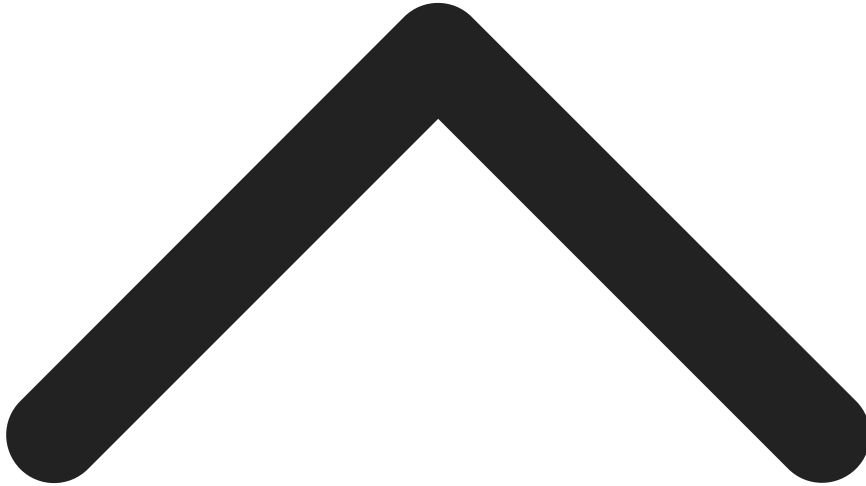
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[Uday Singh](#)

Nov 02, 2017

I was asked to do this in $o(1)$ memory in an interview! Anyone knows how to do that?



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1

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3

5

4

6

```
class Solution {  
    public boolean isValid(String s) {  
  
    }  
  
}
```

```
}
```



Saved

Ln 4, Col 9

You must run your code first



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